Reactions

Aldehydes

(A) Formation

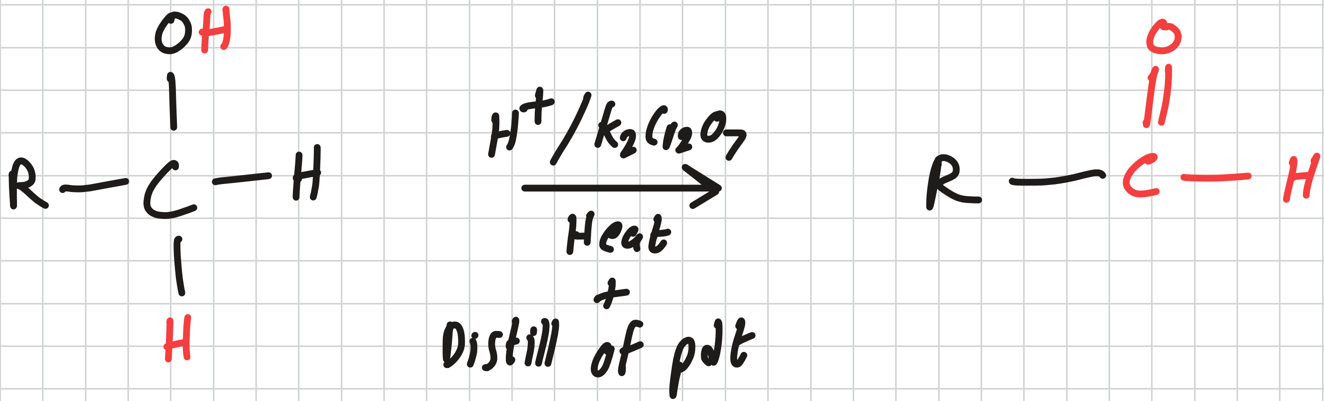
Oxidation of primary alcohol and distillation

Formation of aldehydes

Oxidation of primary alcohols with distillation

Reagents: Acidified $K_2Cr_2O_7$ or Acidified $KMnO_4$

Conditions: Distillation



1° Alcohol

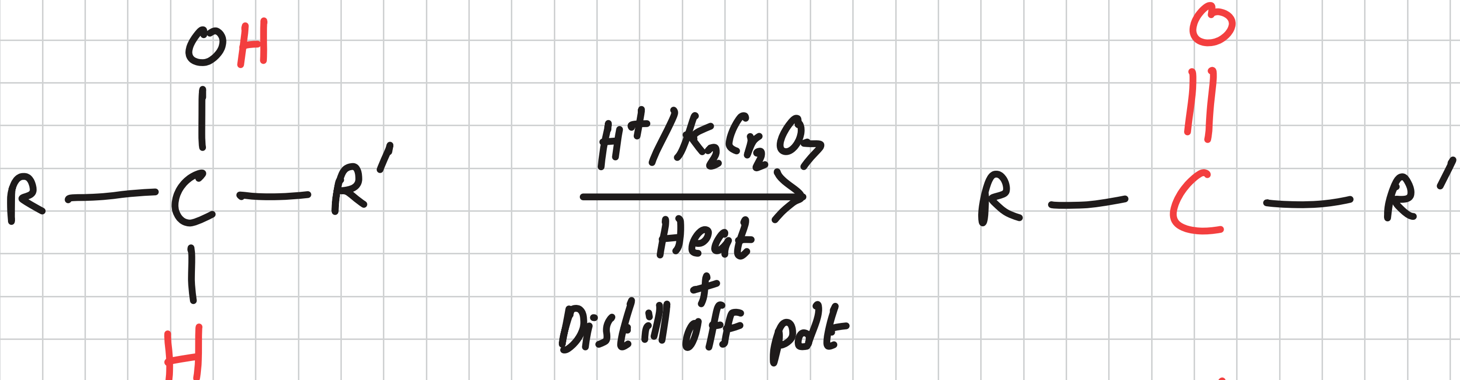
Aldehyde

Formation of ketones

Oxidation of secondary alcohol with distillation

Reagents: Acidified $K_2Cr_2O_7$ or Acidified $KMnO_4$

Conditions: Distillation



2° Alcohol

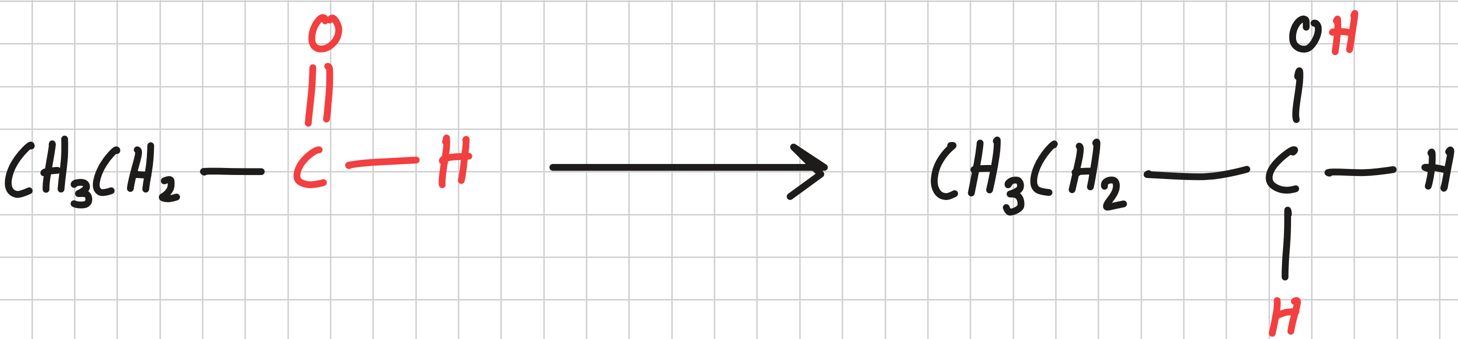
Ketone

Reduction of aldehydes

Aldehydes are reduced to form primary alcohols

Reagents: $NaBH_4$ (aq) or $LiAlH_4$ in dry ether

Conditions: Heat



Propanal

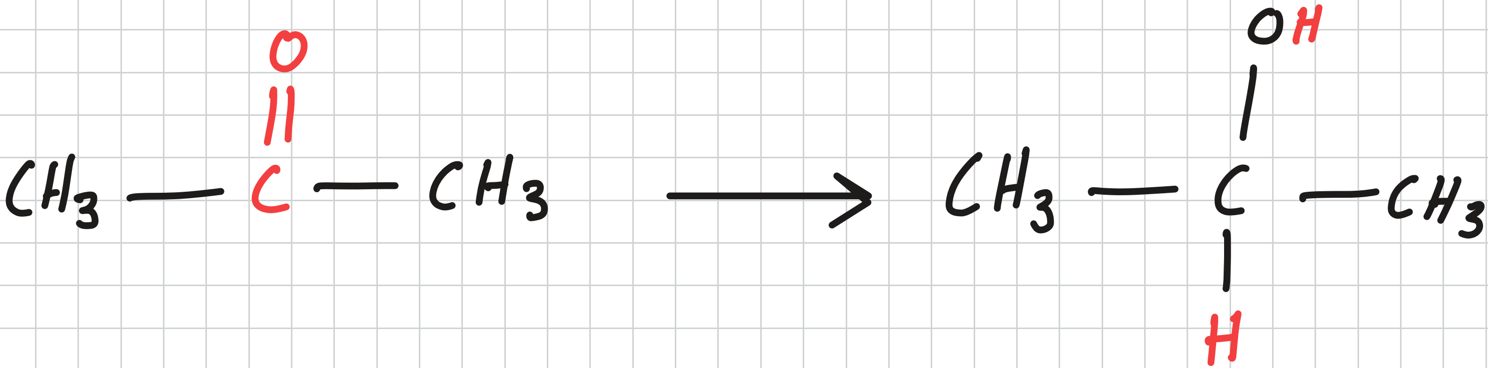
Propan-1-ol

Reduction of Ketones

Ketones are reduced to form secondary alcohols

Reagents: $NaBH_{4(aq)}$ <u>or</u> $LiAlH_4$ in dry ether

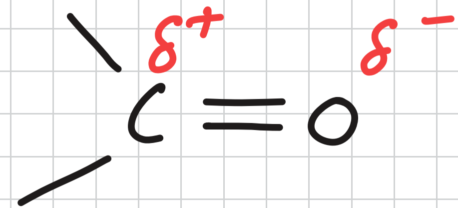
Conditions: Heat



Propanone → Propan-2-ol

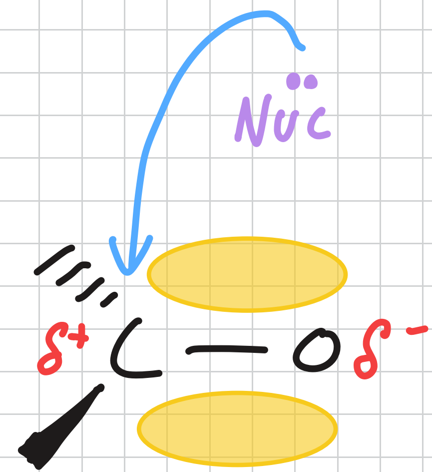
Nucleophilic Addition

The carbonyl group in aldehydes and ketones is polar because oxygen is much more electronegative than hydrogen.



The shared electron in the double bond is drawn towards the oxygen atom. This leaves a partial positive charge on the carbon atom.

As a result, the carbon atom in the carbonyl group is open to attack by nucleophiles.



Both aldehydes and ketones undergo nucleophillic addition.

It involves addition to the polar C = O double bond. Nucleophile'S attack the positive carbon center.

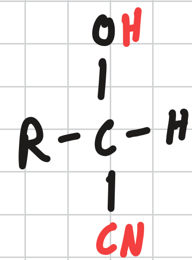
Nucleophilic Addition

Carbonyl compounds add on to HCN to give hydroxynitriles.

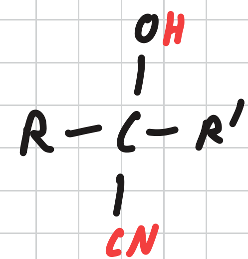
Reagents: HCN with KCN or NaOH cat.

Conditions: Heat under reflux

$$RCHO + HCN \longrightarrow RCH(OH)CN$$



$$RCOR' + HCN \longrightarrow RC(OH)(CN)R'$$



Nucleophilic Addition Mechanism – NaCN cat.

$$CH_3C(=O)-H + HCN \xrightarrow[\text{cat.}]{NaCN} CH_3-C(OH)(CN)-H$$

Step 1: Formation of Nucleophile

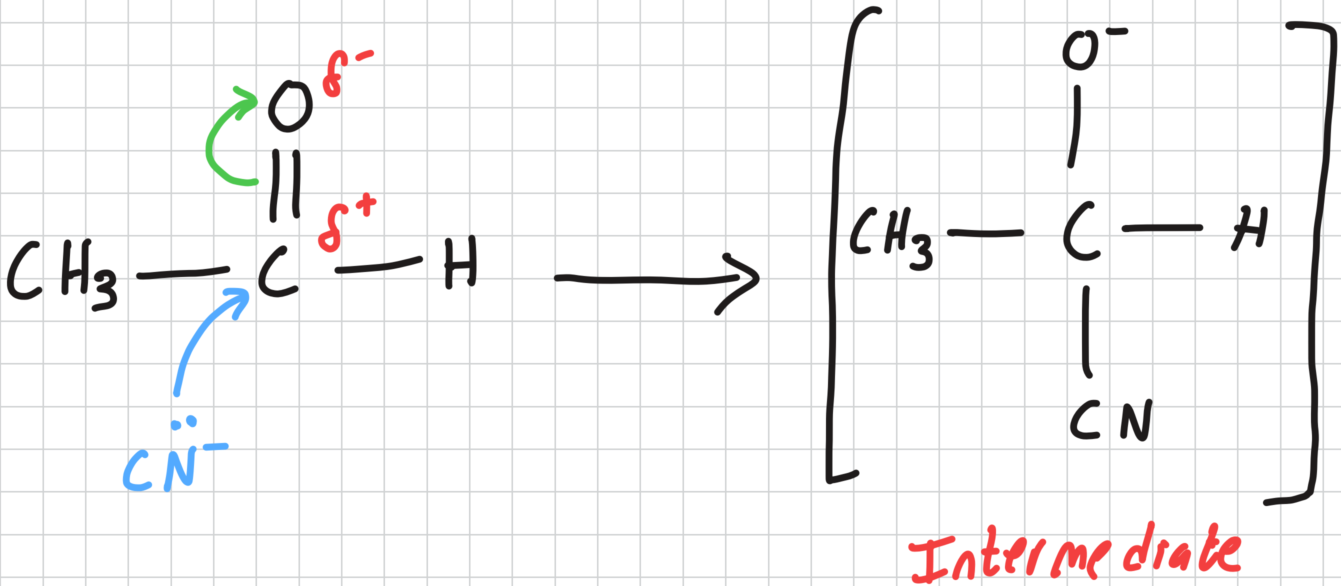
NaCN is a salt. It ionises to form Na^+ and CN^-

$$NaCN \longrightarrow Na^+ + CN^-$$

Nucleophilic Addition Mechanism – NaCN cat.

Step 2: Formation of intermediate

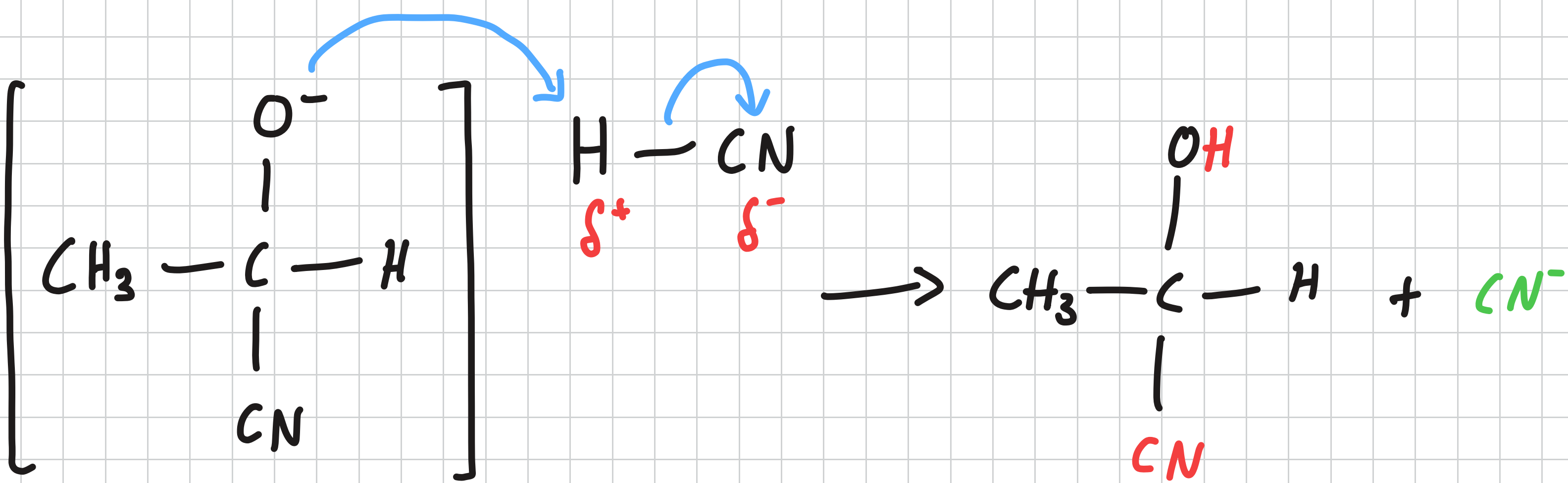
CN⁻ attacks the $C^{\delta+}$ to form an intermediate



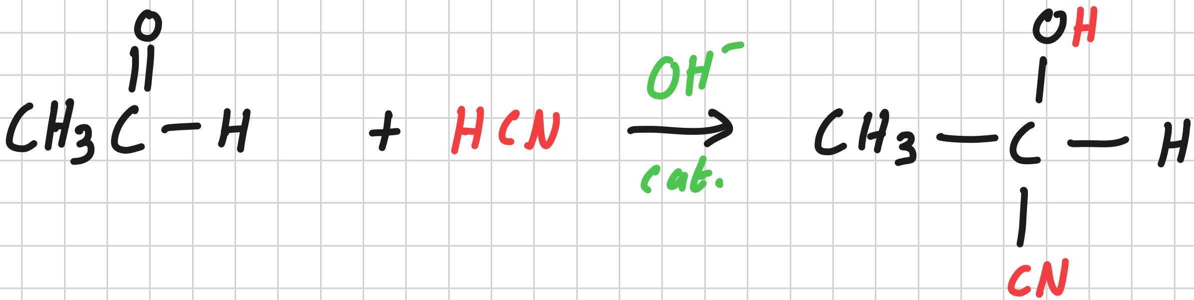
Nucleophilic Addition Mechanism – NaCN cat.

Step 3: Formation of pdt and catalyst regeneration.

The intermediate attacks the HCN to form the product and catalyst is regenerated.



Nucleophilic Addition Mechanism – Alkaline Cat.



Nucleophilic Addition Mechanism — Alkaline Cat.

Step 1: Formation of Nucleophile

HCN partially dissociates, producing CN^-

$$HCN \rightleftharpoons H^+ + CN^-$$

OH^- reacts with H^+, $[H^+]\downarrow$. System reacts according to LCP so as $[H^+]\uparrow$. Forward reaction is favoured.

$\therefore$ More CN^- is formed

$$H^+ + OH^- \longrightarrow H_2O$$

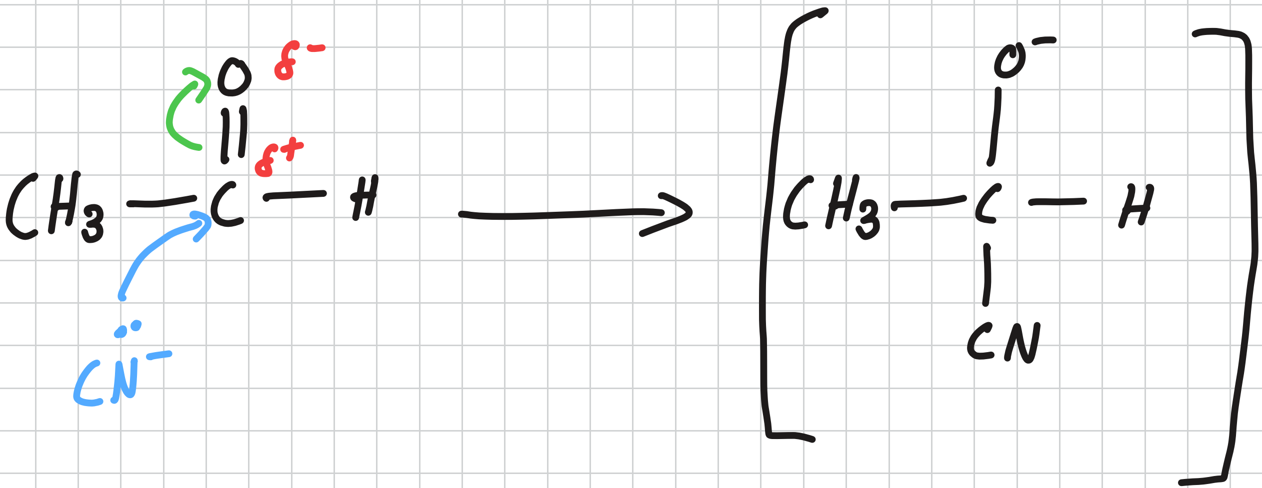
$$HCN \longrightarrow H^+ + CN^-$$

According to LCP

Nucleophilic Addition Mechanism – Alkaline Cat.

Step 2: Formation of intermediate

CN^- attacks $C^{\delta+}$ to form an intermediate

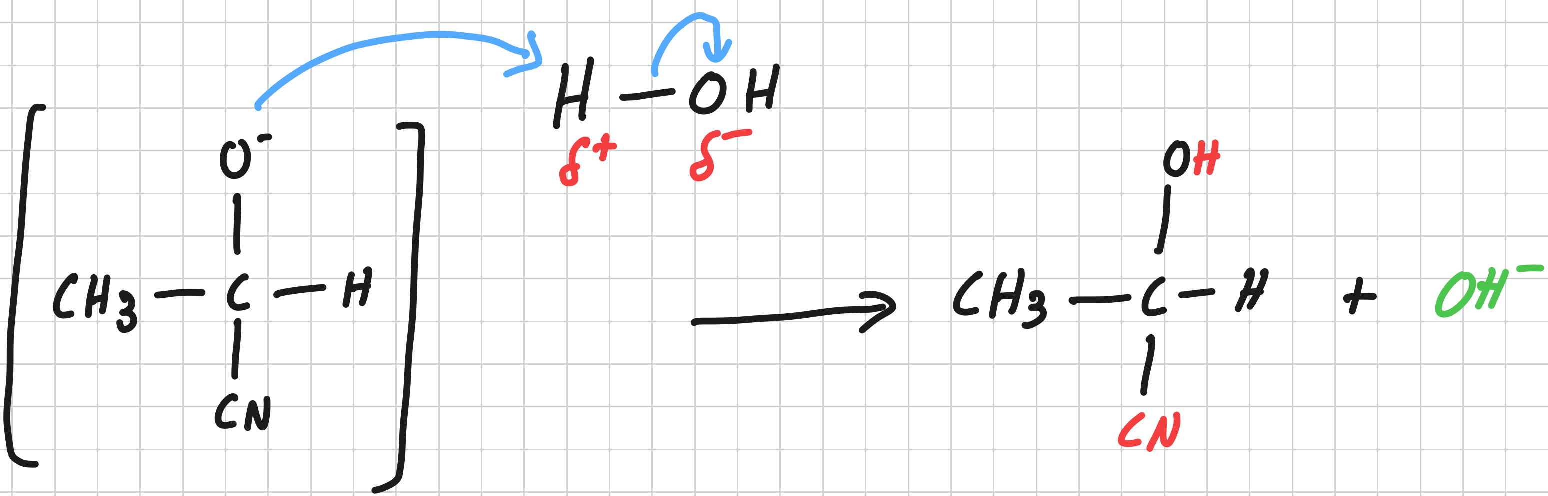


Intermediate

Nucleophilic Addition Mechanism — Alkaline Cat.

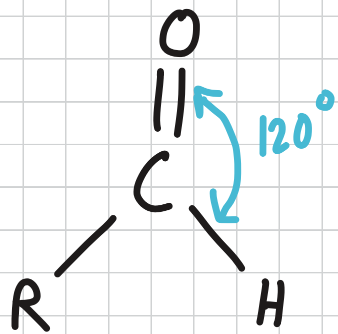
Step 3: Formation of pdt and regeneration of catalyst

The intermediate attacks H_2O to form the pdt and catalyst is regenerated.



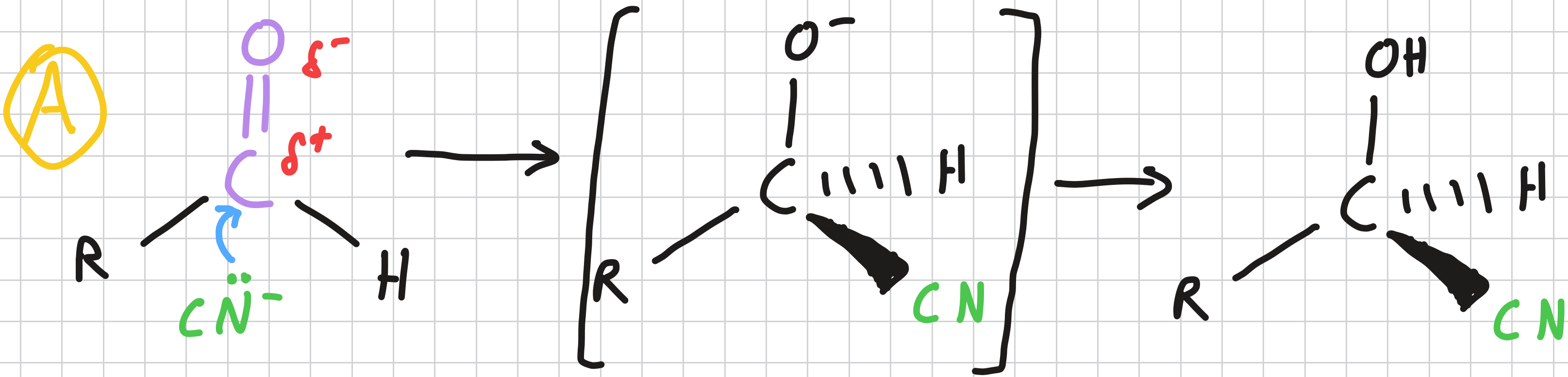
Nucleophilic Addition - Racemic Mixture of Pdts

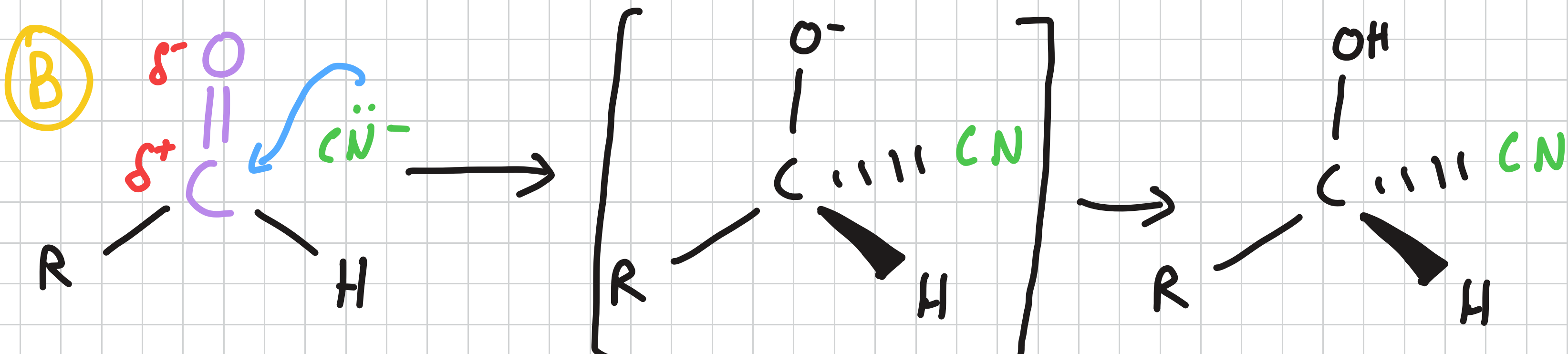
- The carbon in a carbonyl group is sp^2 hybridised.
- The adjacent atoms in the carbonyl group have a trigonal planar shape.
- The carbon atom and its neighbouring atoms all lie in the same plane.



Nucleophilic Addition – Racemic Mixture of Pdts

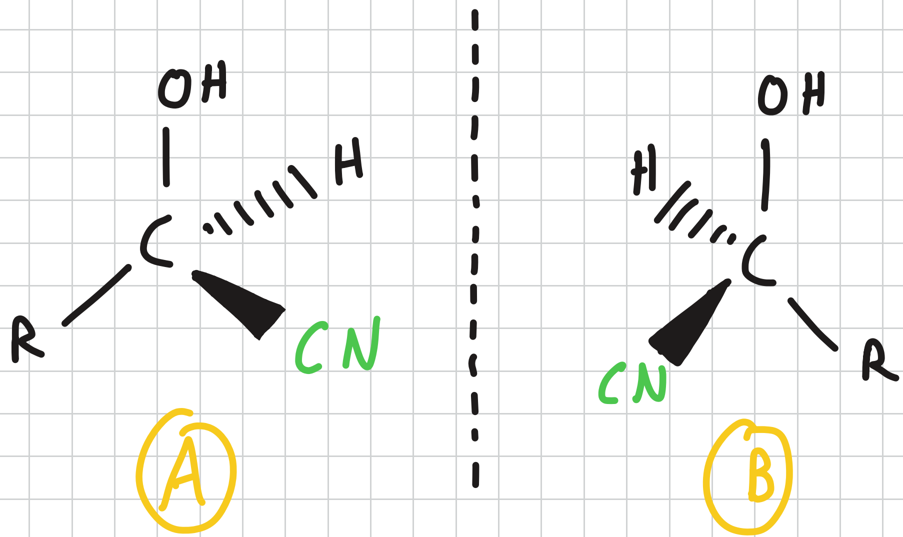
- A CN^- can attack the C=O from either side





Nucleophilic Addition – Racemic Mixture of Pdts

- The products formed when the CN^- attacks from the front or back are **enantiomers** i.e. non superimposable mirror images of each other



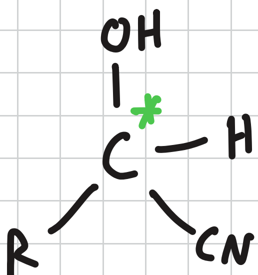
Nucleophilic Addition – Racemic Mixture of Pdts

- Since $C^{\delta+}=O^{\delta-}$ is the same whether CN^- is attacking from the front or back, CN^- is equally likely to attack from front or back.
- This means that half of the products will be one of the enantiomers while the other half will be the other enantiomer.

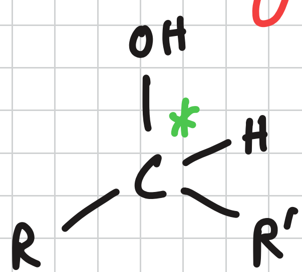
- Hence, the resulting mixture will be racemic, with equimolar amounts of either enantiomer.

Nucleophilic Addition – Racemic Mixture of Pdts

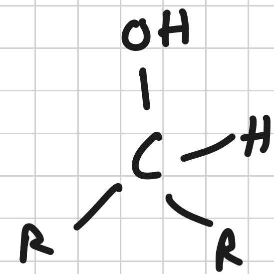
- The product mixture will be racemic only if the product has a chiral carbon.
- All aldehyde derived products will have a C^*.
- Ketone derived products will have a C^* only if the ketone is not symmetric.



From Aldehyde



From Non-Symmetric Ketone



From Symmetric Ketone

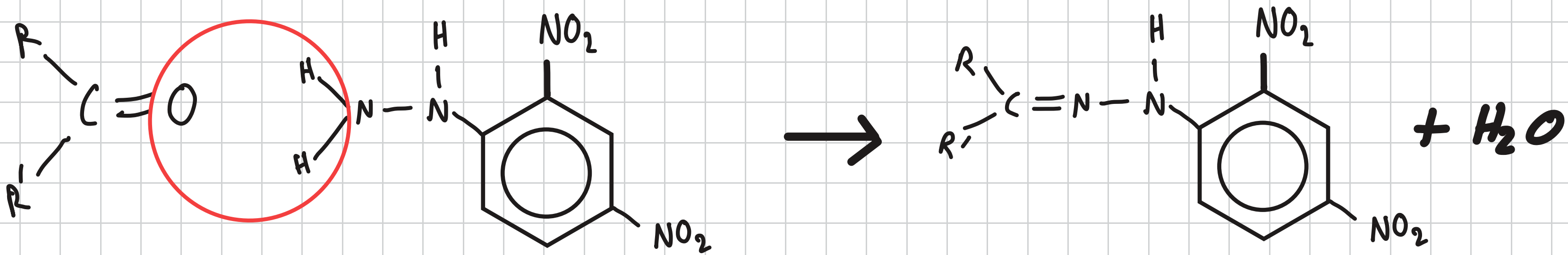
No racemic mixture

2,4-DNPH

2,4-Dinitrophenylhydrazine (2,4-DNPH) is used to test for the presence of carbonyl group. (Aldehyde or ketone)

Reagent: 2,4-DNPH

Observations if carbonyl present: Orange precipitate



Carbonyl 2,4-DNPH Orange ppt

Tollen's Reagent

Tollen's reagent is used to distinguish between aldehydes and ketones.

Reagent: Tollen's reagent

Conditions: Heat

Observations with: Aldehyde: Silver mirror

Ketone : NVC

[O]

$$RCHO + Ag^+ \longrightarrow Ag_{(s)} + RCOO^-$$

[H]

Tollen's Reagent

Fehling's Solution

Fehling's solution is used to distinguish between aldehydes and ketones.

Reagent: Fehling's solution

Observations with: Aldehyde : Brick red precipitate

Ketone : NVC

$$RCHO + Cu^{2+} \longrightarrow Cu_2O + RCOO^-$$

Blue soln

Brick red ppt